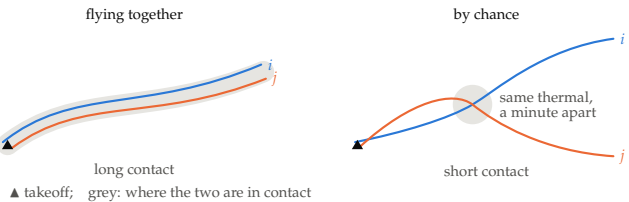
The effect of chance on group identification

Group identificaiton

October 2026

The question

- Contact: close in space, at most δ apart in time.
- Same takeoff, ridges, thermals: chance contacts, mostly short.
- We keep contacts lasting $\geq \tau^*$.



Can a null model that counts the chance contacts give τ^*?

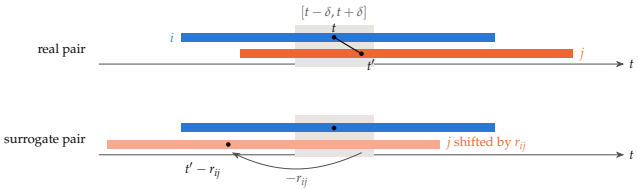
162 485 flights, 6 209 dates, FFVL archive; profiles on 300 dates.

Part 1: Null model, false discovery rate, τ^*

Null model: shift one flight in time

Surrogate pair. i, j same date, $\|\mathbf{x}_i(k_i^0) - \mathbf{x}_j(k_j^0)\| < R_s = 10$ km:

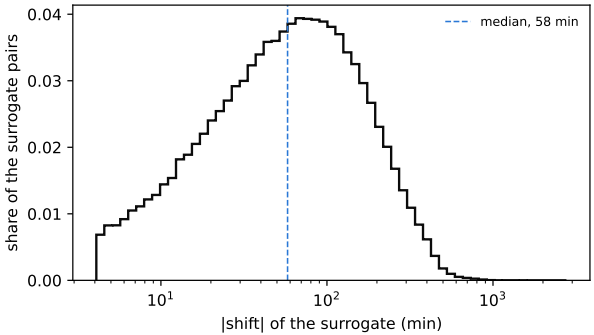
$$\mathbf{x}_j^S(t) = \mathbf{x}_j(t + r), \qquad r = (k_j^0 - k_i^0) - \vartheta, \qquad \vartheta \sim \hat{P}\big(k_j^0 - k_i^0\big), \qquad |r| \geq 2\delta + \Delta$$



$$(t, t') \in \mathcal{M}_{ij} \;\Rightarrow\; |t - (t' - r)| \geq |r| - \delta > \delta$$

Keeps route, terrain, date; destroys every real correspondence. Spiegel et al. (2016).

Shifts of about an hour



248 741 surrogate pairs, whole archive.

$$\text{median } |r| = 58 \text{ min}$$

$$|r| \in [8 \text{ min}, 4.4 \text{ h}] \quad (90\%)$$

$\Rightarrow$ the copy flies in the weather of another hour

False discovery rate

Definitions.

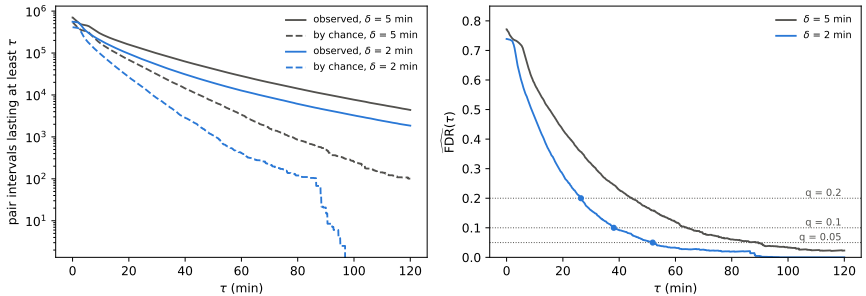
- $M_d(\tau)$: surrogate intervals $\geq \tau$ on date d
- S_d, H_d: surrogate and real exposure on date d (time both fly)
- $N(\tau)$: real intervals $\geq \tau$, all dates

$$\rho_{0,d}(\tau) = \frac{M_d(\tau)}{S_d}, \qquad C(\tau) = \sum_d \rho_{0,d}(\tau)\, H_d$$

$$\widehat{\mathrm{FDR}}(\tau) = \frac{C(\tau)}{N(\tau)}, \qquad \tau^* = \min\{\tau : \widehat{\mathrm{FDR}}(\tau) \leq q\}$$

SAM (Tusher et al. 2001), with time shifts in place of permutations.

Most short contacts are chance



$$\widehat{\mathrm{FDR}}(0) = 0.74, \qquad \widehat{\mathrm{FDR}}(10\ \mathrm{min}) = 0.47, \qquad \widehat{\mathrm{FDR}}(30\ \mathrm{min}) = 0.16$$

$$q = 0.1 \;\Rightarrow\; \tau^* = 38.1\ \mathrm{min}$$

$\delta = 2$ min (blue), 5 min (grey). Bracket: 5–95% over 200 resamplings of the dates.

How much the shift matters

surrogates	$\widehat{\mathrm{FDR}}(10\ \mathrm{min})$	$\widehat{\mathrm{FDR}}(30\ \mathrm{min})$	τ^* (min)
all $\lvert r\rvert$	0.48	0.16	38.2
$\lvert r\rvert < 30$ min	0.73	0.31	53.3
$\lvert r\rvert \geq 60$ min	0.26	0.05	23.0

$\tau^* \in [23, 53]$ min, set by $|r|$

Why τ^* is longer for short $|r|$. A real follower of i more than δ behind is no contact in the real data. A short shift brings it within δ: it counts as chance, though it is a real pair.

Why τ^* is shorter for long $|r|$. Two real pilots of the same hour both go where the thermals work, so they meet without flying together. The copy of j flies in another hour's weather and does not go where i is. These chance meetings are in the real data but not in the surrogates: chance is underestimated.

$q = 0.1$, $\delta = 2$ min, 2 737 dates with all classes. We keep r from the takeoff delays.

Part 2: parameters

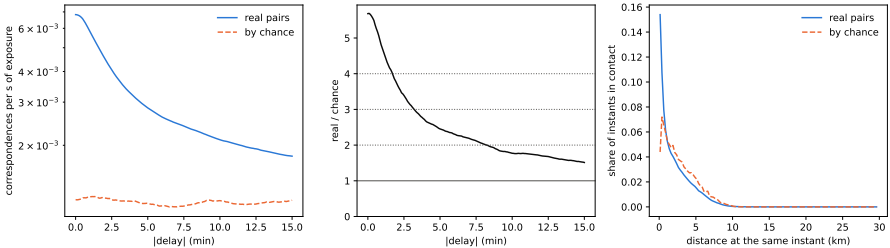
Measured, chosen, or definitions

Three kinds of parameters

	value	set by
δ	2 min	chosen/measured: $R(l)$
q	0.1	chosen/measured
τ^*	38.1 min	$\widehat{\mathrm{FDR}}(\tau^*) \leq q$
ε_h, ε_z, g	500 m, 300 m, 120 s	chosen
$\theta_{\max}$	45°	no effect
Δ, $h_{\max}$, R_{dep}, $v_{\max}$, R_s, $r_{\min}$		fixed: bounds, exactness

δ: delay profile

$$R(l) = \frac{n_{\text{real}}(l)/H_{\text{real}}}{n_{\text{chance}}(l)/H_{\text{chance}}}, \qquad n(l) = \#\{(t, t+l) \in \mathcal{M}\}$$



$$R_{\max} = 5.6, \quad R(105\,\text{s}) \approx 4, \quad R(195\,\text{s}) \approx 3, \quad R(500\,\text{s}) \approx 2, \quad R(15\,\text{min}) = 1.5$$

300 dates, window 15 min. Fast part: flying together; slow tail: same thermals, same hour.

δ: long contacts, short delays

$$\ell_{\text{med}} = \text{median}\{l^*(t) : t \in [t_{\text{on}}, t_{\text{off}}), \text{ no search}\}$$

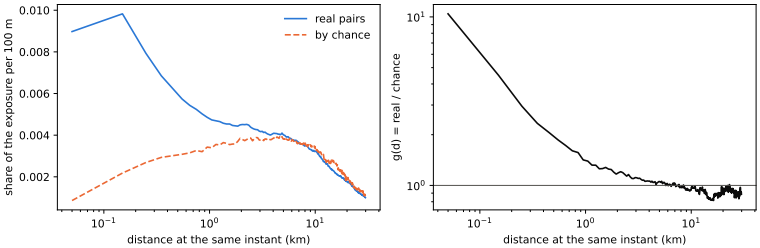
duration	median $\lvert\ell_{\text{med}}\rvert$	$\lvert\ell_{\text{med}}\rvert \le 2$ min	$\lvert\ell_{\text{med}}\rvert \ge 285$ s
< 10 min	210 s	32%	29%
10–30 min	160 s	39%	12%
30–90 min	80 s	64%	2.1%
≥ 90 min	25 s	90%	6 of 10 734

$$\delta = 2 \text{ min}: \qquad R(\delta) = 3.7, \qquad P\big(|\ell_{\text{med}}| \le \delta \;\big|\; \text{duration} \ge 90 \text{ min}\big) = 0.9$$

Run with $\delta = 5$ min, 965 770 intervals.

Distance at the same instant

$$g(d) = \frac{p_{\text{real}}(d)}{p_{\text{chance}}(d)}, \qquad d = \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|$$

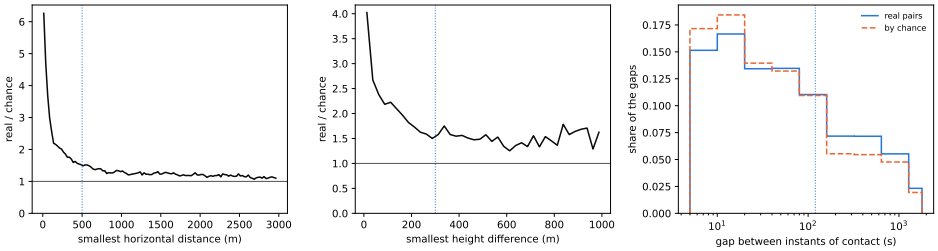


$g(d < 100\text{ m}) = 10.4, \quad g < 1 \text{ for } d > 5.7\text{ km}, \quad \text{in contact: } d \le \varepsilon_h + v_{\max}\delta \approx 3.8\text{ km}$

210 dates, every instant both fly. Excess beyond sight (1.5 km): weather of the hour too.

Tolerances: no edge

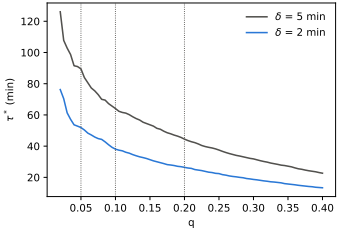
$$d_h(t) = \min_{|l|\le\delta,\ |\Delta z_l|<\varepsilon_z} \|\Delta \mathbf{x}_l\|, \qquad d_z(t) = \min_{|l|\le\delta,\ \|\Delta \mathbf{x}_l\|<\varepsilon_h} |\Delta z_l|$$

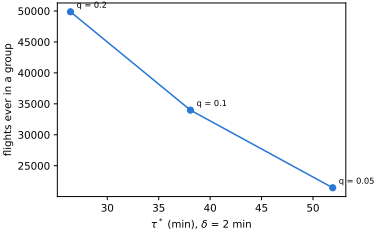


$\dfrac{\text{real}}{\text{chance}} \approx 1.5$ at $\varepsilon_h,\ \varepsilon_z$; gaps: same lengths $\Rightarrow$ $\varepsilon_h, \varepsilon_z, g$ definitions

$\Delta \mathbf{x}_l = \mathbf{x}_i(t) - \mathbf{x}_j(t+l)$, $\Delta z_l = z_i(t) - z_j(t+l)$; 210 dates, $\delta = 2$ min.

q: how much chance we accept





q	τ^* (min)	para in a group	hang in a group	episodes
0.05	51.9	13.7%	3.8%	10 553
0.1	38.1	21.7%	6.9%	17 169
0.2	26.4	31.7%	13.9%	26 504

$\delta = 2$ min. We take $q = 0.1$.

In short

$$\text{chance:} \quad \mathbf{x}_j(t) \to \mathbf{x}_j(t+r), \quad |r| \sim 1\ \text{h}$$

$$\widehat{\text{FDR}}(\tau^*) \leq 0.1 \ \Rightarrow \ \tau^* = 38.1\ \text{min}$$

$$\delta = 2\ \text{min}: \quad \text{real/chance}(\delta) = R(\delta) = 3.7, \quad 90\%\ \text{of long contacts}$$

$$\varepsilon_h,\ \varepsilon_z,\ g: \ \text{definitions}$$

In a group at some point: 21.7% of the paraglider flights, 6.9% of the hang-glider flights; 17 169 episodes.